
Malware Code Injection

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Agenda

- Overview of malware code injection
- Different kinds of code injection technique and their usage in malware
- What do investigators look at in code injections
- How to detect

Overview

- When we run malware such as Dridex payload, Brazilian malware in the lab, nothing seems happen. Actually, behind view, there is code injection took place.

In this presentation,

- we will review the basic idea of code injection
- see some examples of malware code injection
- known why it is stealthy
- What do investigators look at in code injections
- Learn different kinds of code injections and their usage in malware
 - DLL injection
 - Process replacement (RunPE)
 - Hook injection
 - APC injection

Examples of code injection

- CASE0541 - .Net IL injector
- CASE9208 – Brazilian malware (RunPE malware)
- CASE0353 – Powershell Spam
- CASE0194 – Dridex payload
- Bangladesh bank got stolen \$8M in Swift bank payment system
- Independent penetration tester announced to “audit” bank software

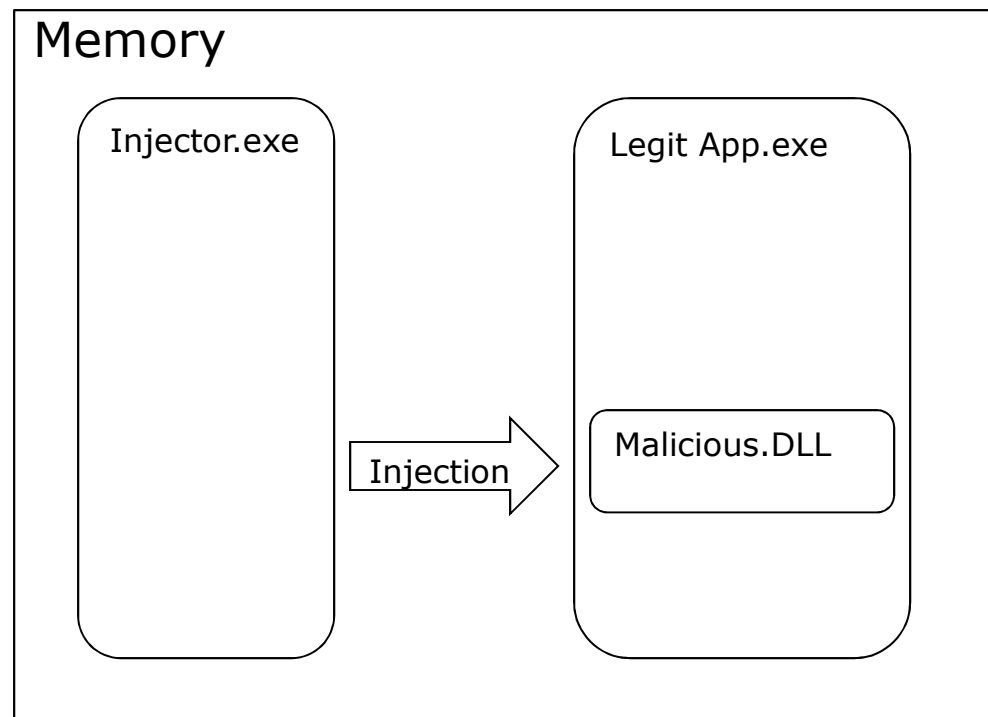
DLL Injection

- DLL Injection

DLL Injection

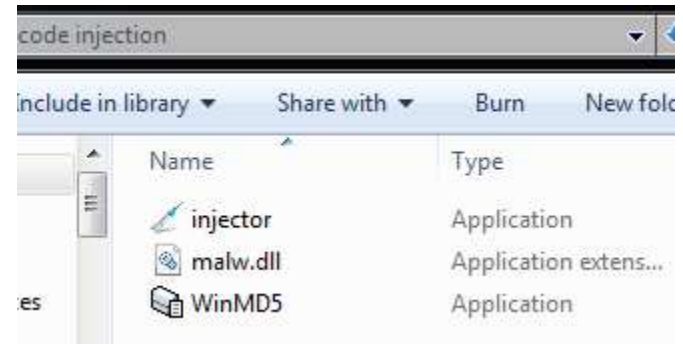
- DLL injection is a way that malware can run a DLL inside another process, therefore providing that DLL with all the privileges of the target process.
- It is a common way to force a process to load malicious code.

- Steps:



Dll Injection 2 - Demo

- 3 files involved:
 - Injector.exe – malware injector
 - Malw.dll – malicious DLL
 - WinMD5.exe – legit app



- Legit app is running in memory (WinMD5.exe)
- We run malware injector.exe with parameter "Malw.dll", "WinMD5.exe".
- Result: Malw.dll is injected to WinMD5.exe and starts execution.

Dll Injection 3 - Demo before dll injection

The screenshot displays the Windows Task Manager window in the background, showing a list of running processes. In the foreground, the WinMD5Free v1.20 application is open, showing a file selection dialog. The application window has a title bar that reads "WinMD5Free v1.20" and a logo that says "WinMD5Free" with the website "www.winmd5.com". The main area of the application is titled "Select a file to compute MD5 checksum (or drag and drop a file onto this window)". It contains a text input field for the file name, a "Browse .." button, and a "Cancel" button. Below this, it shows "File Name and Size: n/a" and "Current file MD5 checksum value: n/a". There is also a section for "Original file MD5 checksum value (optional)" with a text input field and a "Verify" button. At the bottom, there are buttons for "Website", "About", and "Exit".

Overlaid on the right side of the Task Manager window is the "WinMD5.exe:12700 Properties" dialog box. The "Threads" tab is selected, showing a table with columns: TID, CPU, Cycles Delta, and Start Address. The table contains one entry: TID 13368, CPU 0.11, Cycles Delta 91,756 K, and Start Address 81,720 K. A red arrow points from the text "Before injection, there is one thread for WinMD5.exe" to the TID 13368 entry. Below the table, there are buttons for "Stack" and "Module". At the bottom of the dialog, there are buttons for "Permissions", "Kill", "Suspend", "OK", and "Cancel".

Process Name	Private Bytes	Working Set	PID	Description	Company Name
msdtc.exe	3,332 K	352 K	1904	Microsoft Distributed Transa...	Microsoft Corporation
taskhost.exe	7,544 K	3,684 K	1988	Host Process for Windows T...	Microsoft Corporation
SearchIndexer.exe	32,104 K	12,220 K	2280	Microsoft Windows Search I...	Microsoft Corporation
svchost.exe	1,476 K	528 K	2956	Host Process for Win...	
svchost.exe	66,788 K	5,968 K	3052	Host Process for Win...	
taskhost.exe	9,108 K	6,940 K	5500		
svchost.exe	2,072 K	3,064 K	13464	Host Process for Win...	
lsass.exe	3,812 K	4,224 K	520	Local Security Author...	
lsn.exe	2,096 K	1,304 K	532		
winlogon.exe	2,400 K	964 K	464		
explorer.exe	0.11	91,756 K	81,720 K	18932 Windows Explorer	
procexp.exe	2,184 K	1,864 K	20136	Sysinternals Process...	
procexp64.exe	3.37	27,172 K	39,296 K	20224 Sysinternals Process...	
WinMD5.exe	2,860 K	8,600 K	12700	WinMD5Free	

WinMD5Free v1.20

WinMD5Free

www.winmd5.com

Select a file to compute MD5 checksum (or drag and drop a file onto this window)

File Name and Size: n/a

Current file MD5 checksum value: n/a

Original file MD5 checksum value (optional). It usually can be found from website or .md5 file

paste its original md5 value to verify

Website About Exit

WinMD5.exe:12700 Properties

Threads

Count: 1

TID	CPU	Cycles Delta	Start Address
13368	0.11	91,756 K	81,720 K

Before injection, there is one thread for WinMD5.exe

Thread ID: 13368

Start Time: 11:26:32 PM 6/19/2016

State: Wait:WrUserRequest

Kernel Time: 0:00:00.359

User Time: 0:00:00.156

Context Switches: 4,519

Cycles: 3,064,365,211

Base Priority: 8

Dynamic Priority: 10

I/O Priority: Normal

Memory Priority: 5

Ideal Processor: 0

Stack Module

Permissions Kill Suspend

OK Cancel

DLL Injection 4 – demo after dll injection

The screenshot displays a Windows task manager window showing various running processes. A red box highlights the 'WinMD5.exe' process, which is running at 0.28 seconds. Below the task manager, the 'WinMD5Free v1.20' application window is visible, showing a file selection interface. To the right, the 'WinMD5.exe:12700 Properties' window is open, showing the 'Threads' tab. A red arrow points to the '14576' thread, with the annotation 'After injection, there is one additional thread'. Further right, the 'Stack for thread 14576' window is open, showing a list of loaded modules. A red box highlights the 'malw.dll+0x1041' entry in the stack, with the annotation 'Stack for thread shows malw.dll'. Above the stack window, a small 'DLL' message box displays 'Hello from DllMain!' with an 'OK' button. A red arrow points to this message box with the annotation 'DllMain in malw.dll is executed. Hello message is showed only for demo purpose'.

Task Manager Processes:

Process Name	Private Bytes	Working Set	PID	Description	Company Name
svchost.exe	< 0.01	21,152 K	15,780	Host Process for Windows S...	Microsoft Corporation
svchost.exe	6,232 K	4,880 K	300	Host Process for Windows S...	Microsoft Corporation
svchost.exe	< 0.01	20,556 K	5,592	712 Host Process for Windows S...	Microsoft Corporation
spoolsv.exe	7,016 K	1,776 K	1068	Spooler SubSystem App	Microsoft Corporation
svchost.exe	7,880 K	3,456 K	1096	Host Process for Windows S...	Microsoft Corporation
svchost.exe	1,760 K	788 K	1284	Host Process for Windows S...	Microsoft Corporation
VGAuthService.exe	5,488 K	280 K	1316	VMware Guest Authenticatio...	VMware, Inc.
vmtoolsd.exe	0.05	11,604 K	6,636	1408 VMware Tools Core Service	VMware, Inc.
msdtc.exe	3,332 K	352 K	1904	Microsoft Distributed Transa...	Microsoft Corporation
taskhost.exe	7,544 K	3,688 K	1988	Host Process for Windows T...	Microsoft Corporation
SearchIndexer.exe	0.01	32,444 K	12,952	2280 Microsoft Windows Search I...	Microsoft Corporation
svchost.exe	1,476 K	528 K	2956	Host Process for Windows S...	Microsoft Corporation
svchost.exe	66,788 K	12,152 K	3052	Host Process for Windows S...	Microsoft Corporation
taskhost.exe	9,108 K	6,940 K	5500	Host Process for Windows T...	Microsoft Corporation
svchost.exe	2,072 K	3,064 K	13464	Host Process for Windows S...	Microsoft Corporation
lsass.exe	3,812 K	4,316 K	520	Local Security Authority	Microsoft Corporation
lsmdc.exe	2,092 K	1,300 K	532	Local Security Authority	Microsoft Corporation
winlogon.exe	2,400 K	960 K	464	Local Security Authority	Microsoft Corporation
explorer.exe	0.02	93,896 K	84,100	18932 Windows Explorer	Microsoft Corporation
procexp.exe	2,184 K	1,900 K	20136	Sysinternals Process Explorer	Sysinternals
procexp64.exe	0.56	31,468 K	44,412	20224 Sysinternals Process Explorer	Sysinternals
WinMD5.exe	0.28	2,952 K	9,164	12700 WinMD5Free	WinMD5
injector.exe	5,648 K	17,364 K	16068	WinMD5Free	WinMD5
vmtoolsd.exe	0.09	17,796 K	12,488	2084 VMware Tools Core Service	VMware, Inc.
procexp.exe	Susp...	6,856 K	1,036	2468 Sysinternals Process Explorer	Sysinternals
010Editor.exe	52,936 K	57,216 K	14,648	010 Editor - The Professional	010

WinMD5.exe:12700 Properties - Threads:

TID	CPU	Cycles Delta	Start Address
13368			WinMD5.exe+0x66768
14576			kernel32.dll!LoadLibraryA

Stack for thread 14576:

```
2 wow64.dll!Wow64SystemServiceEx+0x1ce
3 wow64.dll!Wow64LdrpInitialize+0x429
4 ntdll.dll!RtlIsDosDeviceName_U+0x24c87
5 ntdll.dll!LdrInitializeThunk+0xe
6 USER32.dll!WaitMessage+0x15
7 USER32.dll!DialogBoxIndirectParamAorW+0x108
8 USER32.dll!SoftModalMessageBox+0x757
9 USER32.dll!SoftModalMessageBox+0xa33
10 USER32.dll!MessageBoxTimeoutW+0x52
11 USER32.dll!MessageBoxTimeoutA+0x76
12 USER32.dll!MessageBoxExA+0x1b
13 USER32.dll!MessageBoxA+0x18
14 malw.dll+0x1041
15 kernel32.dll!ReflectiveLoader@A+0x5ffa
```

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- DLL Injection calls APIs
- Pseudocode of Injector.exe:

```
hTargetProcess = OpenProcess(,,TargetProcessID);  
pNameProc=VirtualAllocEx(hTargetProcess,,)  
WriteProcessMemory(hTargetProcess,,maliciousDLLName,,)  
GetModule("Kernel32.dll")  
GetProcAddress(..,"LoadLibraryA");  
CreateRemoteThread(hTargetProcessm,,,,pNameProc,,)
```

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IDA Pro View of injector.exe

```
inc    eax
cmp    cl, bl
jnz    short loc_403C50
```

```
sub    eax, edx
lea    ebp, [eax+1]
mov    eax, [esp+3Ch+dwProcessId]
push   eax                ; dwProcessId
push   ebx                ; binheritHandle
push   1F0FFFh            ; dwDesiredAccess
call   ds:OpenProcess
mov    esi, eax
cmp    esi, ebx
jz     loc_403D7C
```

1, Attach to target process.
OpenProcess to get handle to the target process

```
push   edi
push   4                  ; flProtect
push   1000h              ; flAllocationType
push   ebp                ; dwSize
push   ebx                ; lpAddress
push   esi                ; hProcess
call   ds:VirtualAllocEx
mov    edi, eax
cmp    edi, ebx
jz     loc_403D7B
```

2, Allocate memory within the target process.
VirtualAllocEx to allocate space for the malicious DLL

```
mov    edx, [esp+40h+lpBuffer]
lea    ecx, [esp+40h+NumberOfBytesWritten]
push   ecx                ; lpNumberOfBytesWritten
push   ebp                ; nSize
push   edx                ; lpBuffer
push   edi                ; lpBaseAddress
push   esi                ; hProcess
call   ds:WriteProcessMemory
test   eax, eax
jz     loc_403D7B
```

3, Copy the dll or the dll path into the process memory and determine appropriate memory address.
WriteProcessMemory to write the name of the malicious DLL into target process

```
mov    al, 45h
mov    [esp+40h+var_13], al
```

Graph overview

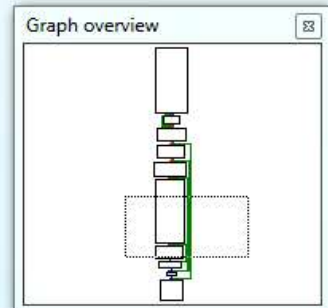
DLL Injection 7

IDA Pro View of injector.exe

```
lea     eax, [esp+40h+ThreadId]
push    eax                ; lpThreadId
mov     cl, 61h
push    ebx                ; dwCreationFlags
mov     [esp+48h+var_22], cl
mov     [esp+48h+var_1C], cl
push    edi                ; lpParameter
lea     ecx, [esp+4Ch+ProcName]
push    ecx                ; lpProcName
lea     edx, [esp+50h+ModuleName]
push    edx                ; lpModuleName
mov     [esp+54h+ModuleName], 48h
mov     [esp+54h+var_12], 52h
mov     [esp+54h+var_11], 4Eh
mov     [esp+54h+var_E], 33h
mov     [esp+54h+var_D], 32h
mov     [esp+54h+var_C], 2Eh
mov     [esp+54h+var_B], 44h
mov     [esp+54h+var_8], b1
mov     [esp+54h+var_23], 6Fh
mov     [esp+54h+var_21], 64h
mov     [esp+54h+var_1F], 69h
mov     [esp+54h+var_1E], 62h
mov     [esp+54h+var_1A], 79h
mov     [esp+54h+var_19], 41h
mov     [esp+54h+var_18], b1
call    ds:GetModuleHandleA
push    eax                ; hModule
call    ds:GetProcAddress
push    eax                ; lpStartAddress
push    ebx                ; dwStackSize
push    ebx                ; lpThreadAttributes
push    esi                ; hProcess
call    ds:CreateRemoteThread
mov     ebp, eax
cmp     ebp, ebx
jz      short loc_403D74
```

```
push    0FFFFFFFh          ; dwMilliseconds
push    ebp                ; hHandle
call    ds:WaitForSingleObject
push    8000h              ; dwFreeType
push    ebx                ; dwSize
push    edi                ; lpAddress
push    esi                ; hProcess
call    ds:VirtualFreeEx
mov     eax, 1
```

4, Instruct the process to execute the dll by calling CreateRemoteThread, and pass the control to DllMain



DLL Injection 8

- In Dll injection, the malware injector never calls a malicious function. The DllMain in the dll is called automatically the dll is loaded into memory.
- The goal of dll injection injector is to call CreateRemoteThread so that a thread is created in the target process.
- Once we find the Dll injection activity in disassembly, we should start looking for strings containing the names of the malicious DLL and the victim process.
- the starter normally get victim PID by comparing the process's name, for this case, it is WinMD5.exe
- To find the malicious dll name, we could set a breakpoint at WriteProcessMemory, and check the stack memory to find the value of Buffer that passed to WriteProcessMemory.

■ Process Replacement

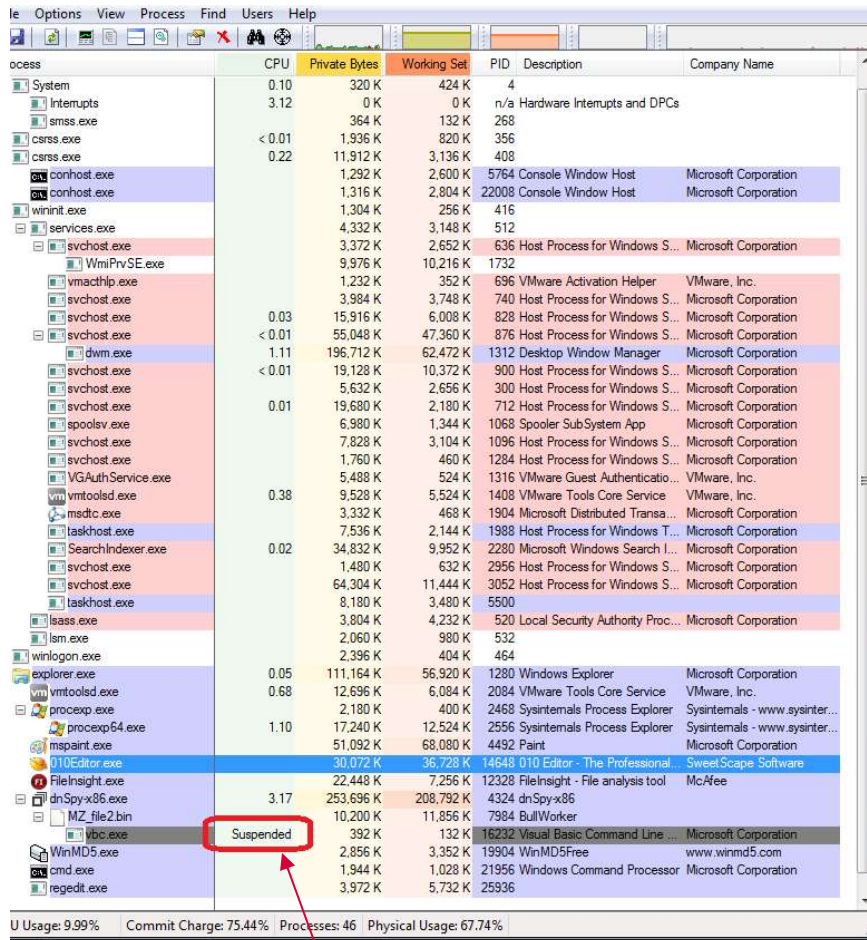
Process replacement (RunPE)

- Malware load a process into memory with a SUSPENDED state.
- Replace the process's memory with malicious executable.
- Resume the process to run the malicious executable

- Malware appears non-malicious by disguising as the normal windows process.
- Bypass firewall, IPS.
- Example: CASE9208, CASE0541
- Key API calls:
 - ZwUnmapViewOfSection
 - ResumeThread

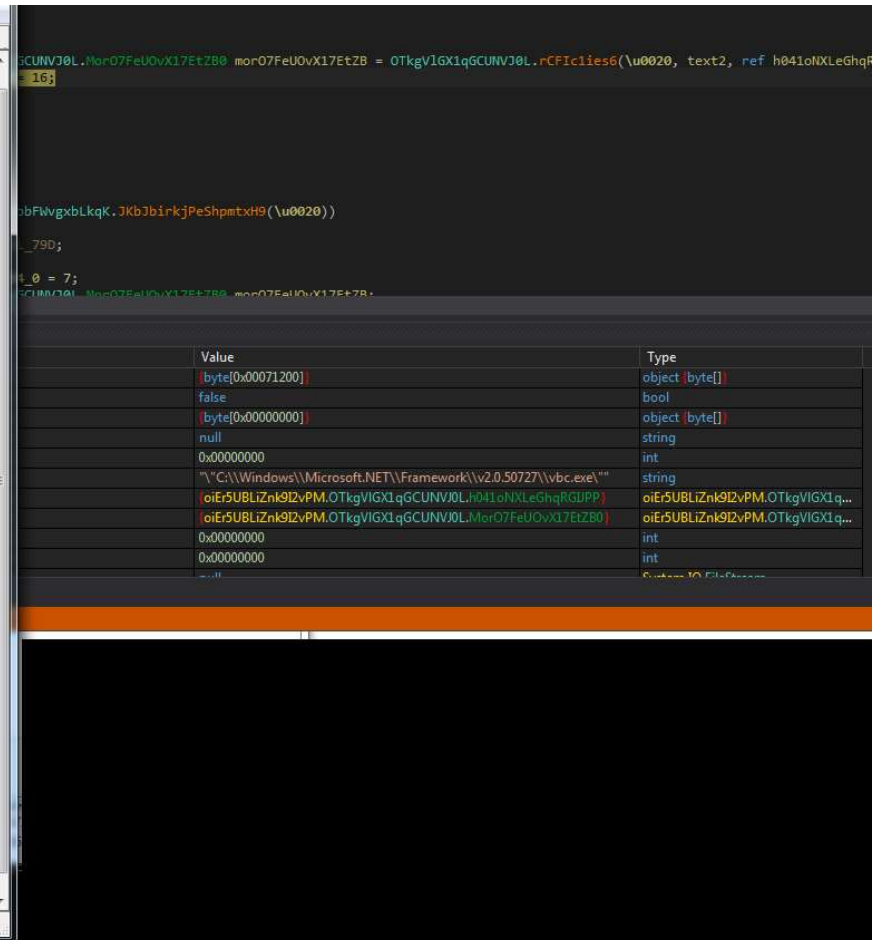
- Detection: compare the PE header of memory with the legit file on disk

Process replacement (RunPE)



Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
System	0.10	320 K	424 K	4		
smss.exe	3.12	0 K	0 K	n/a	Hardware Interrupts and DPCs	
csrss.exe		364 K	132 K	268		
csrss.exe	< 0.01	1,936 K	820 K	356		
csrss.exe	0.22	11,912 K	3,136 K	408		
conhost.exe		1,292 K	2,600 K	5764	Console Window Host	Microsoft Corporation
conhost.exe		1,316 K	2,804 K	22008	Console Window Host	Microsoft Corporation
wininit.exe		1,304 K	256 K	416		
services.exe		4,332 K	3,148 K	512		
svchost.exe		3,372 K	2,652 K	636	Host Process for Windows S...	Microsoft Corporation
WmiPrvSE.exe		9,976 K	10,216 K	1732		
vmacthlp.exe		1,232 K	352 K	696	VMware Activation Helper	VMware, Inc.
svchost.exe		3,984 K	3,748 K	740	Host Process for Windows S...	Microsoft Corporation
svchost.exe	0.03	15,916 K	6,008 K	828	Host Process for Windows S...	Microsoft Corporation
svchost.exe	< 0.01	55,048 K	47,360 K	876	Host Process for Windows S...	Microsoft Corporation
dwf.exe	1.11	196,712 K	62,472 K	1312	Desktop Window Manager	Microsoft Corporation
svchost.exe	< 0.01	19,128 K	10,372 K	900	Host Process for Windows S...	Microsoft Corporation
svchost.exe		5,632 K	2,656 K	300	Host Process for Windows S...	Microsoft Corporation
svchost.exe	0.01	19,680 K	2,180 K	712	Host Process for Windows S...	Microsoft Corporation
spoolsv.exe		6,980 K	1,344 K	1068	Spooler Sub System App	Microsoft Corporation
svchost.exe		7,828 K	3,104 K	1096	Host Process for Windows S...	Microsoft Corporation
svchost.exe		1,760 K	460 K	1284	Host Process for Windows S...	Microsoft Corporation
VGAuthService.exe		5,488 K	524 K	1316	VMware Guest Authenticatio...	VMware, Inc.
vmtoolsd.exe	0.38	9,528 K	5,524 K	1408	VMware Tools Core Service	VMware, Inc.
msdtc.exe		3,332 K	468 K	1904	Microsoft Distributed Transa...	Microsoft Corporation
taskhost.exe		7,536 K	2,144 K	1988	Host Process for Windows T...	Microsoft Corporation
SearchIndexer.exe	0.02	34,832 K	9,952 K	2280	Microsoft Windows Search I...	Microsoft Corporation
svchost.exe		1,480 K	632 K	2956	Host Process for Windows S...	Microsoft Corporation
svchost.exe		64,304 K	11,444 K	3052	Host Process for Windows S...	Microsoft Corporation
taskhost.exe		8,180 K	3,480 K	5500		
lsass.exe		3,804 K	4,232 K	520	Local Security Authority Proc...	Microsoft Corporation
ls.exe		2,060 K	980 K	532		
winlogon.exe		2,396 K	404 K	464		
explorer.exe	0.05	111,164 K	56,920 K	1280	Windows Explorer	Microsoft Corporation
vmtoolsd.exe	0.68	12,696 K	6,084 K	2084	VMware Tools Core Service	VMware, Inc.
proccp.exe		2,180 K	400 K	2468	Sysinternals Process Explorer	Sysinternals - www.sysinter...
proccp64.exe	1.10	17,240 K	12,524 K	2556	Sysinternals Process Explorer	Sysinternals - www.sysinter...
mspaint.exe		51,092 K	68,080 K	4492	Paint	Microsoft Corporation
010Editor.exe		30,072 K	36,728 K	14648	010 Editor - The Professional...	SweetScape Software
FileInSight.exe		22,448 K	7,256 K	12328	FileInSight - File analysis tool	McAfee
dnSpy-x86.exe	3.17	253,696 K	208,792 K	4324	dnSpy-x86	
MZ_file2bin		10,200 K	11,856 K	7984	BullWorker	
vbc.exe		392 K	132 K	16232	Visual Basic Command Line ...	Microsoft Corporation
WinMD5.exe		2,856 K	3,352 K	19904	WinMD5Free	www.winmd5.com
cmd.exe		1,944 K	1,028 K	21956	Windows Command Processor	Microsoft Corporation
regedit.exe		3,972 K	5,732 K	25936		

U Usage: 9.99% Commit Charge: 75.44% Processes: 46 Physical Usage: 67.74%



```
GCUNVJ0L.Mor07FeU0vX17EtZB0 mor07FeU0vX17EtZB = OTkgVIG1q6CUNVJ0L.rCFic1ies6(\u0020, text2, ref h041oNXLeGhqR
= 16;

bFwvgxbLkqK.3Kb3birkjPeShpmtxH9(\u0020))

_790;

4 0 = 7;
CUMQ0L.Mor07FeU0vX17EtZB0 mor07FeU0vX17EtZB =
```

Value	Type
byte[0x00071200]	object [byte[]]
false	bool
byte[0x00000000]	object [byte[]]
null	string
0x00000000	int
"C:\Windows\Microsoft.NET\Framework\v2.0.50727\vbc.exe"	string
oiEr5UBLiZnK9I2vPM.OTkgVIG1q6CUNVJ0L.h041oNXLeGhqRGUPP	oiEr5UBLiZnK9I2vPM.OTkgVIG1q...
oiEr5UBLiZnK9I2vPM.OTkgVIG1q6CUNVJ0L.Mor07FeU0vX17EtZB0	oiEr5UBLiZnK9I2vPM.OTkgVIG1q...
0x00000000	int
0x00000000	int

Malware loads process in Suspended status
C:\Windows\Microsoft.Net\Framework\v2.0.50727\vbc.exe

Process replacement (RunPE)

```
.Kv7bFUadEAqcoxRgpkM(array, f4aQMkR7NSa9x0TmuB.AZ7176aTN1TjyFcJHcC(f-
00 66 00 77 00 65 00 .....X.....$.#.$3.2.f.d.f.w.e.f.w.e.
00 65 00 66 00 77 00 f.w.e.....H+.....$.#.$3.2.f.d.f.w.e.f.w.
00 00 00 00 00 00 00 e.f.w.e.....Q+.....$#@32fdfwefwefwe.....
00 1D 00 00 00 1C 00 X.....n.t.d.l.l...d.l.l.....X.....
00 44 00 46 00 42 00 ..a.l.c.V.S.l.5.T.F.j.I.P.F.h.T.p.F.T.Y.D.F.B
4F 66 53 65 63 74 69 F.N.T.C.4.=.....Q+...NtUnmapViewOfSecti
00 66 00 77 00 65 00 on....X.....$.#.$3.2.f.d.f.w.e.f.w.e.
00 65 00 66 00 77 00 f.w.e.....H+.....$.#.$3.2.f.d.f.w.e.f.w.
00 00 00 00 00 00 00 e.f.w.e.....Q+.....$#@32fdfwefwefwe.....
00 65 00 63 00 74 00 X.....N.t.U.n.m.a.p.V.i.e.w.O.f.S.e.c.t.
A6 6A 02 00 00 00 00 i.o.n.....+.....|.....|.....j.....
00 53 00 6C 00 5A 00 l...p.#w.....X.....T.0.Y.y.S.l.Z.
5E 55 56 48 13 09 0A e.V.V.Z.I.E.w.k.K.....Q+.....OF2JV^UVH...
00 00 00 00 00 00 00 .....
```

Malware calls NtUnmapViewOfSection to replace the memory of legit process with malicious code

Hook Injections

- Hook Injection intercepts messages destined for applications.
- SetWindowsHookEx is used to intercept Windows messages
- Often seen in Keyloggers

APC Injections

- APC (Asynchronous procedure call) is OS feature that allows user application and OS components to execute code in a thread. APCs are processed when a thread calls functions like Sleep, WaitForSingleObjectEx.
- Investigators look for QueueUserAPC API calls
- Often seen in RootKit and malware drivers

What do Malware analysts look at code injection

- Malware analysts identify injection techniques in order to find malware on a live system.
- Find the target process by tracing the malware calling to APIs. Breakpoints may be needed at APIs.
- Find the malicious DLL by identify important strings in memory.

Impacts to security controls - bypassing

- Code injection is stealthy and dangerous.
- By injecting to legit IE, malware could gain communicate to C2 server, bypassing Firewall
- By injecting to legit application in the memory, malware could bypass hash based AV detection, Bit9, or IPS.
- By injecting to Windows process, malware could gain system privilege
- Malware could hide from viewing in a process list

Detection and remediation

- Review API calls
- Compare PE file in memory and on disk
- Memory forensic, volatility

Malware code injection

- Questions
- Thank you!